

Exercise 3 Fundamentals Corrections

Q1

```
SELECT product_name,
       price,
CASE WHEN price > 1000 THEN 'Expensive'
      WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
      ELSE 'Budget'
END AS price_category
FROM products;
```

product_name	price	price_category
Laptop	1200	Expensive
Phone	800	Mid-range
Keyboard	45	Budget
Monitor	300	Mid-range
Mouse	25	Budget

Q2

```
SELECT customer_name,
       amount,
CASE WHEN amount >= 1000 THEN 'High Value'
      WHEN amount BETWEEN 500 AND 999.99 THEN 'Medium Value'
      ELSE 'Low Value'
END AS order_value_category
FROM orders;
```

customer_name	amount	order_value_category
Alice	150,00	Low Value
Bob	500,00	Medium Value
Charlie	999,99	Medium Value
Diana	45,50	Low Value
Ethan	1200,00	High Value

③ SELECT emp_name,
department,
salary,

CASE WHEN department = 'IT' AND salary > 80000 THEN 'Senior IT'

WHEN department = 'HR' AND salary > 55000 THEN 'Experienced HR'

ELSE 'Staff'

END AS position_level

FROM employees;

emp_name	department	salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	Experienced HR
Mark	IT	75000	Staff
Lucy	Finance	85000	Staff
Tom	HR	55000	Staff

④ SELECT student_name,
score,

CASE WHEN score >= 90 THEN 'A'

WHEN score BETWEEN 80 AND 89 THEN 'B'

WHEN score BETWEEN 70 AND 79 THEN 'C'

WHEN score BETWEEN 60 AND 69 THEN 'D'

ELSE 'F'

END AS grade

FROM students;

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

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⑤

```

SELECT delivery_id,
       delivery_time_minutes,
CASE WHEN delivery_time_minutes <= 30 THEN 'Fast'
      WHEN delivery_time_minutes BETWEEN 31 AND 60 THEN 'On Time'
      ELSE 'Late'
END AS performance
FROM deliveries;

```

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

```

⑥ SELECT issue_type,
       priority,
CASE WHEN days priority = 3 THEN 'High'
      WHEN priority = 2 THEN 'Medium'
      WHEN priority = 1 THEN 'Low'
END AS priority_label
FROM tickets;

```

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

⑦

```

SELECT student_id,
(days-present * 100 / total-days) AS attendance-percentage,
CASE WHEN (days-present * 100 / total-days) >= 90 THEN 'Excellent'
      WHEN (days-present * 100 / total-days) BETWEEN 75 AND 89 THEN 'Good'
      ELSE 'Need Improvement'
END AS attendance-status
FROM attendance;
  
```

Student id	attendance-percentage	Attendance status
1	90.0	Excellent
2	60.0	Need Improvement
3	96.0	Excellent
4	50.0	Need Improvement
5	100	Excellent

⑧ SELECT product-id,
stock-qty,

```

CASE WHEN stock-qty = 0 THEN 'Out of Stock'
      WHEN stock-qty BETWEEN 1 AND 4 THEN 'Low Stock'
      ELSE 'In Stock'
END AS stock-status
FROM products_inventory;
  
```

product-id	Stock-qty	Stock status
1	5	In stock
2	0	Out of Stock
3	25	In stock
4	10	In stock
5	2	Low Stock

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```
SELECT subject,
        enrolled_students,
        CASE enrolled_students > 25 THEN 'Large'
        WHEN enrolled_students BETWEEN 10 AND 24 THEN 'Medium'
        ELSE 'Small'
        END AS class_size_category
FROM classes;
```

subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

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```
SELECT payment_id,
        payment_method,
        amount,
        CASE WHEN payment_method = 'Cash' AND amount >= 200 THEN 'Eligible for Discount'
        ELSE 'Not Eligible'
        END AS discount_eligibility
FROM payments;
```

payment_id	payment_method	amount	discount_eligibility
1	Card	50.00	Not Eligible
2	Cash	200.00	Eligible for Discount
3	Card	150.00	Not Eligible
4	Paypal	75.00	Not Eligible
5	Cash	300.00	Eligible for Discount